NS-3 Quick Cheat Sheet

1.,Create Nodes

```
NodeContainer nodes;
nodes.Create(n);
```

2. Connect Nodes

Point-to-point

```
PointToPointHelper p2p;
p2p.SetDeviceAttribute("DataRate", StringValue("10Mbps"));
p2p.SetChannelAttribute("Delay", StringValue("2ms"));

NetDeviceContainer devices;
devices = p2p.Install(nodes);
```

3. Wi-Fi

```
WifiHelper wifi;
WifiMacHelper mac;
YansWifiPhyHelper phy;
YansWifiChannelHelper channel =
   YansWifiChannelHelper::Default();

phy.SetChannel(channel.Create());

wifi.SetRemoteStationManager(
   "ns3::ConstantRateWifiManager");

NetDeviceContainer devices =
   wifi.Install(phy, mac, nodes);
```

4. Install Internet Stack

```
InternetStackHelper internet;
internet.Install(nodes);
```

5. Assign IP Addresses

```
Ipv4AddressHelper address;

address.SetBase("10.1.1.0",
                "255.255.255.0");

Ipv4InterfaceContainer interfaces =
   address.Assign(devices);
```

6. UDP Echo Application

Server

```
UdpEchoServerHelper server(9);

ApplicationContainer serverApp =
   server.Install(nodes.Get(1));

serverApp.Start(Seconds(1.0));
serverApp.Stop(Seconds(10.0));
```

Client

```
UdpEchoClientHelper client(
   interfaces.GetAddress(1), 9);

client.SetAttribute("MaxPackets",
                    UintegerValue(10));

client.SetAttribute("Interval",
                    TimeValue(Seconds(1.0)));

client.SetAttribute("PacketSize",
                    UintegerValue(1024));

ApplicationContainer clientApp =
   client.Install(nodes.Get(0));
```

7. Mobility

Constant Position

```
MobilityHelper mobility;

mobility.SetMobilityModel(
   "ns3::ConstantPositionMobilityModel");

mobility.Install(nodes);
```

8. Random Mobility

```
mobility.SetMobilityModel(
   "ns3::RandomWalk2dMobilityModel",
   "Bounds",
   RectangleValue(
      Rectangle(-50, 50, -50, 50)));
```

9. **AODV**

```
AodvHelper aodv;

InternetStackHelper internet;
internet.SetRoutingHelper(aodv);
internet.Install(nodes);
```

10. Logging

Enable logging:

```
LogComponentEnable(
   "UdpEchoClientApplication",
   LOG_LEVEL_INFO);
```

11. Command-Line Arguments

```
uint32_t nNodes = 5;
```

```
CommandLine cmd;
cmd.AddValue("nNodes",
             "Number of nodes",
             nNodes);
cmd.Parse(argc, argv);
```

12. PCAP Tracing

```
p2p.EnablePcapAll("simulation")
```

13. FlowMonitor

```
#include "ns3/flow-monitor-module.h"
```

Create monitor:

```
FlowMonitorHelper flowmon;

Ptr<FlowMonitor> monitor =
   flowmon.InstallAll();
```